Generic small-signal silicon diode · ILLUSTRATIVE REFERENCE

ILLUSTRATIVE TEACHING DATA. This is not a manufacturer datasheet. Values are rounded, typical figures for a generic small-signal silicon switching diode, provided so CircuitSense can demonstrate PDF retrieval with page citations. Always use the datasheet of the exact part you have.

1. Overview

A small-signal silicon diode conducts when its anode is more positive than its cathode by roughly the forward voltage. It is used for signal rectification, reverse-polarity protection of low-current circuits, clamping and logic OR-ing.

2. Pin identification

The cathode is marked by a band on the body. In KiCad's Device:D symbol pin 1 is the cathode (K) and pin 2 is the anode (A).

Electrical characteristics (typical, illustrative)

3. Electrical characteristics

Forward voltage: about 0.6 V to 0.7 V at 1 mA, rising to about 1 V at 100 mA (illustrative).

Maximum continuous forward current: 150 mA (illustrative).

Maximum repetitive reverse voltage: 75 V (illustrative).

Reverse leakage current: a few nanoamps at 25 °C, increasing strongly with temperature.

4. Troubleshooting

Diode test on a multimeter (power removed): about 0.5 V to 0.7 V forward, open-circuit (OL) in reverse. A reading near 0 V in both directions indicates a shorted diode; OL in both directions indicates an open diode.